Instruction Mapping

High Nibble		Low Nibble 3210 F 1111	E 1110	D 1101	C 1100	B 1011	A 1010	9 1001	8 1000	7 0111	6 0110	5 0101	4 0100	3 0011	2 0010	1 0001	0 0000
7654																	
1111	F	HALT					BR <d>	BNC <d>	BRC <d>	BNZ <d>	BRZ <d>	JMP	JEQ	JNC	JC	JNZ	JZ
1110	E															LDPC	STPC
1101	D														RETI	RET	CALL
1100	C																
1011	B														BTST	BCLR	BSET
1010	A													RLC	RRC	INC	DEC
1001	9									ROLC	RORC	SHLC	SHRC	ROL	ROR	SHL	SHR
1000	8													AND	OR	XOR	NOT
0111	7												CMP	MUL	DIV	SUB	ADD
0110	6									SSF	CSF	SOF	COF	SZF	SCF	CZF	CCF
0101	5													LDFL	LDI	LDM	LD
0100	4														MOV	XFER	ST
0011	3														RSP	STSP	LDSP
0010	2													POPF	PUSHF	POP	PUSH
0001	1	OUT R7	OUT R6	OUT R5	OUT R4	OUT R3	OUT R2	OUT R1	OUT R0	IN R7	IN R6	IN R5	IN R4	IN R3	IN R2	IN R1	IN R0
0000	0													DI	EI	INT	NOP

PC Flags

76 <- Bits

11 Jump/Call Instructions (+ HALT)

10 Register Specific ALU Operations

01 Register Moves involving Memory/Immediate values/ALU Operations

00 Misc Instructions and Stack Operations

LDLW/LDHW - 32Bit CPU only.

	Type		Instruction Bits							Source Register				Destination Register			
	T	T	i	i	i	i	i	i	0	Rs	Rs	Rs	0	Rd	Rd	Rd	
0x00	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	NOP
0x01	0	0	0	0	0	0	0	1	0	0	0	0	0	I	I	I	INT x
0x02	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	EI
0x03	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	DI
0x1x	0	0	0	1	0	Rd	Rd	Rd	P	P	P	P	P	P	P	P	IN <Port>
0x1x	0	0	0	1	1	Rs	Rs	Rs	P	P	P	P	P	P	P	P	OUT <Port>
0x20	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	PUSH
0x21	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	POP
0x22	0	0	1	0	0	0	1	0	0	0	0	0	O	N	C	Z	PUSHF
0x23	0	0	1	0	0	0	1	1	0	0	0	0	O	N	C	Z	POPF
0x30	0	0	1	1	0	0	0	0	0	0	0	0	0	Rd	Rd	Rd	LDSP <Rd>
0x31	0	0	1	1	0	0	0	1	0	Rs	Rs	Rs	0	0	0	0	STSP <Rs>
0x32	0	0	1	1	0	0	1	0	0	0	0	0	0	0	0	0	RSP
0x40	0	1	0	1	0	0	0	0	0	Rs	Rs	Rs	0	0	0	0	ST
0x41	0	1	0	0	0	0	0	1	0	Rs	Rs	Rs	0	Rd	Rd	Rd	XFER
0x42	0	1	0	0	0	0	1	0	0	0	0	0	0	Rd	Rd	Rd	MOV
0x50	0	1	0	1	0	0	0	0	0	0	0	0	0	Rd	Rd	Rd	LD
0x51	0	1	0	1	0	0	0	1	0	0	0	0	0	Rd	Rd	Rd	LDM
0x52	0	1	0	1	0	0	1	0	0	0	0	0	0	Rd	Rd	Rd	LDI
0x53	0	1	0	1	0	0	1	1	0	0	0	0	0	Rd	Rd	Rd	LDFL
0X60	0	1	1	0	0	0	0	0	0	Rs	Rs	Rs	0	0	0	0	PUSH
0X61	0	1	1	0	0	0	0	1	0	0	0	0	0	Rd	Rd	Rd	POP
0x70	0	1	1	1	0	0	0	0	0	Rs	Rs	Rs	0	Rd	Rd	Rd	ADD
0x71	0	1	1	1	0	0	0	1	0	Rs	Rs	Rs	0	Rd	Rd	Rd	SUB
0x72	0	1	1	1	0	0	1	0	0	Rs	Rs	Rs	0	Rd	Rd	Rd	DIV
0x73	0	1	1	1	0	0	1	1	0	Rs	Rs	Rs	0	Rd	Rd	Rd	MUL
0x74	0	1	1	1	0	1	0	0	0	Rs	Rs	Rs	0	Rd	Rd	Rd	CMP
0X80	1	0	0	0	0	0	0	0	0	Rs	Rs	Rs	0	0	0	0	NOT
0X81	1	0	0	0	0	0	0	1	0	Rs	Rs	Rs	0	0	0	0	XOR
0X82	1	0	0	0	0	0	1	0	0	Rs	Rs	Rs	0	0	0	0	OR
0X83	1	0	0	0	0	0	1	1	0	Rs	Rs	Rs	0	0	0	0	AND
0X90	1	0	0	1	0	0	1	0	0	Rs	Rs	Rs	0	0	0	0	SHR
0X91	1	0	0	1	0	0	1	0	0	Rs	Rs	Rs	0	0	0	0	SHL
0X92	1	0	0	1	0	0	1	0	0	Rs	Rs	Rs	0	0	0	0	ROR
0X93	1	0	0	1	0	0	1	0	0	Rs	Rs	Rs	0	0	0	0	ROL
0X94	1	0	0	1	0	0	1	0	0	Rs	Rs	Rs	0	0	0	0	SHRC
0X95	1	0	0	1	0	0	1	0	0	Rs	Rs	Rs	0	0	0	0	SHLC
0X96	1	0	0	1	0	0	1	0	0	Rs	Rs	Rs	0	0	0	0	RORC
0X97	1	0	0	1	0	0	1	0	0	Rs	Rs	Rs	0	0	0	0	ROLC
0xA0	1	0	1	0	0	0	0	0	0	0	0	0	0	Rd	Rd	Rd	DEC
0xA1	1	0	1	0	0	0	0	1	0	0	0	0	0	Rd	Rd	Rd	INC
0xA2	1	0	1	0	0	0	1	0	0	0	0	0	0	Rd	Rd	Rd	RRC
0xA3	1	0	1	0	0	0	1	1	0	0	0	0	0	Rd	Rd	Rd	RLC
0xB0	1	0	1	1	0	0	0	0	b3	b2	b1	b0	0	Rd	Rd	Rd	BSET
0xB1	1	0	1	1	0	0	0	1	b3	b2	b1	b0	0	Rd	Rd	Rd	BCLR
0xB2	1	0	1	1	0	0	1	0	b3	b2	b1	b0	0	Rd	Rd	Rd	BTST
0xD0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	CALL
0xD1	1	1	0	1	0	0	0	1	0	0	0	0	0	0	0	0	RET
0xD2	1	1	0	1	0	0	1	0	0	0	0	0	0	0	0	0	RETI
0xE0	1	1	1	0	0	0	0	0	0	0	0	0	0	Rd	Rd	Rd	STPC
0xE1	1	1	1	0	0	0	0	1	0	0	0	0	0	Rd	Rd	Rd	LDPC
0xF0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	JZ
0xF1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	JNZ
0xF2	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	JC
0xF3	1	1	1	1	0	0	1	1	0	0	0	0	0	0	0	0	JNC
0xF4	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	JNE
0xF5	1	1	1	1	0	1	0	1	0	0	0	0	0	0	0	0	JMP
0xF6	1	1	1	1	0	1	1	0	d	d	d	d	d	d	d	d	BRZ <disp>
0xF7	1	1	1	1	0	1	1	0	d	d	d	d	d	d	d	d	BNZ <disp>
0xF8	1	1	1	1	1	0	0	0	d	d	d	d	d	d	d	d	BRC <disp>
0xF9	1	1	1	1	1	0	0	1	d	d	d	d	d	d	d	d	BNC <disp>
0xFA	1	1	1	1	1	0	1	0	d	d	d	d	d	d	d	d	BR <disp>